PROGRAMMING IN JAVA – WEEK 7, ASSIGNMENT 7

Q1. What will be the output of the following Java program?

```
import java.io.*;
class Public {
   public static void main(String[] args) throws IOException {
      FileReader fr = new FileReader("NPTEL.txt");
      BufferedReader br = new BufferedReader(fr);
      System.out.println(br.readLine());
      br.close();
   }}
```

Options:

a. Hello, World!

b. This is Programming in Java online course.

c. IOException

d. null

Correct Answer: b. This is Programming in Java online course.

Q2. Complete the following snippet with the required code: File file = new File("file.txt"); if (_____) { System.out.println("File exists."); } else { System.out.println("File does not exist."); }

```
File file = new File("file.txt");
if (_________________) {
   System.out.println("File exists.");
} else {
   System.out.println("File does not exist.");
}
```

Options:

a. file.exists()

b. file.isFile()

c. file.fileExists()

d. file.isAvailable()

Correct Answer: a. file.exists()

Q3. What is the output of the following Java program? final int i; i = 20; i = 30; System.out.println(i);

```
class Main {
   public static void main(String args[]) {
      final int i;
      i = 20;
      i = 30;
      System.out.println(i);
   }
}
```

Options:

a. 30

b. Compiler Error

c. Garbage value

d. 0

Correct Answer: b. Compiler Error

Q4. Which value is returned by read() when EOF is encountered?

Options:

a. 0

b. 1

c. -1

d. Null

Correct Answer: c. -1

Q5. What will be the output of the Calculator program where num = 100 and calc(2) sets this.num = num * 10?

```
public class Calculator {
   int num = 100;
   public void calc(int num) {
      this.num = num * 10;
   }
   public void printNum() {
      System.out.println(num);
   }
   public static void main(String[] args) {
      Calculator obj = new Calculator();
      obj.calc(2);
      obj.printNum();
   }
}
```

Options:

a. 20

b. 100

c. 1000

d. 2

Correct Answer: a. 20

Q6. What is the primary difference between Byte Streams and Character Streams in Java?

Options:

a. Byte Streams handle characters, while Character Streams handle bytes.

b. Byte Streams are used for binary data, while Character Streams are used for image data.

c. Character Streams are faster than Byte Streams in all cases.

d. Byte Streams are used for binary data, while Character Streams are used for text data.

Correct Answer: d. Byte Streams are used for binary data, while Character Streams are used for text data.

Q7. What is the primary purpose of thread synchronization in Java?

Options:

a. To allow multiple threads to execute a method at the same time.

b. To ensure thread execution follows a specific order.

c. To ensure data inconsistency.

d. To prevent race conditions and ensure data consistency.

Correct Answer: d. To prevent race conditions and ensure data consistency.

Q8. Which stream is used to read data from a source such as a file, array, peripheral device, or socket?

Options:

a. InputStream

b. OutputStream

c. Input/OutputStream

d. DataOutputStream

Correct Answer: a. InputStream

Q9. Which class in Java is used to create a new directory?

Options:

a. FileReader

b. FileWriter

c. File

d. Directory

Correct Answer: c. File

Q10. What is the primary purpose of input streams in Java?

Options:

a. To write data to a file.

b. To read data from a file.

c. To delete a file.

d. To create directories.

Correct Answer: b. To read data from a file.

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WEEK 8 — ASSIGNMENT 8

QUESTIONS, OPTIONS AND CORRECT ANSWERS

Q1. Which of the following is TRUE about check boxes and radio buttons?

a. Check boxes allow only one selection, while radio buttons allow multiple selections.

b. Check boxes allow multiple selections, while radio buttons allow only one selection.

c. Both allow only one selection.

d. Both allow multiple selections.

Correct Answer: b

Q2. Which layout manager arranges components in a rectangular grid?

a. FlowLayout

b. CardLayout

c. BorderLayout

d. GridLayout

Correct Answer: d

Q3. Which method is used to set the title of a Frame in Java?

a. setTitle()

b. setName()

c. setText()

d. setLabel()

Correct Answer: a

Q4. Which method is used to request that a component be repainted in Java?

a. paint()

b. repaint()

c. update()

d. draw()

Correct Answer: b

Q5. Which AWT component is used to display a scrolling list of items?

a. Label

b. List

c. Button

d. TextField

Correct Answer: b

Q6. Which layout manager arranges components in a flow, placing them one after another?

a. FlowLayout

b. BorderLayout

c. GridLayout

d. CardLayout

Correct Answer: a

Q7. What does "AWT" stand for in Java?

a. Abstract Widget Toolkit

b. Advanced Window Toolkit

c. Abstract Window Toolkit

d. Advanced Widget Toolkit

Correct Answer: c

Q8. Which AWT component is used to create a button in a GUI?

a. Label

b. Button

c. TextField

d. Checkbox

Correct Answer: b

Q9. Which of the following correctly describes the GUI shown in the assignment?

a. It contains two Labels and one TextField.

b. It contains one Frame, two Labels, and two TextFields.

c. It contains two Frames, two Labels, and two TextFields.

d. It contains one Frame, two Buttons, and two TextFields.

Correct Answer: b

Q10. Which of the following is TRUE about a check box in Java?

a. A check box cannot be in the on or off state.

b. Clicking a check box changes its state between on and off.

c. A check box can be on and off at the same time.

d. A check box cannot be grouped with other check boxes.

Correct Answer: b

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ASSIGNMENT 10

QUESTIONS, OPTIONS AND CORRECT ANSWERS

Q1. Which Swing component is used to display a window?

a. JFrame

b. JButton

c. JLabel

d. JTextField

Correct Answer: a. JFrame

Q2. Which Swing component is used to enter a single line of text?

a. JButton

b. JLabel

c. JTextField

d. JPanel

Correct Answer: c. JTextField

Q3. Which Swing component is commonly used to display non-editable text?

a. JLabel

b. JButton

c. JFrame

d. JTextField

Correct Answer: a. JLabel

Q4. Which JDBC interface is used to execute SQL statements?

a. ResultSet

b. Statement

c. Connection

d. DriverManager

Correct Answer: b. Statement

Q5. Which JDBC object contains the result of a SELECT query?

a. Connection

b. Statement

c. ResultSet

d. DriverManager

Correct Answer: c. ResultSet

Q6. Which JDBC class is used to establish a database connection?

a. DriverManager

b. ResultSet

c. Statement

d. Scanner

Correct Answer: a. DriverManager

Q7. Which Java class is used to connect a client to a server?

a. ServerSocket

b. Socket

c. URL

d. NetworkInterface

Correct Answer: b. Socket

Q8. Which class is used by a server to listen for client connections?

a. Socket

b. URL

c. ServerSocket

d. NetworkInterface

Correct Answer: c. ServerSocket

Q9. Which protocol is used in the URL https://nptel.ac.in/?

a. FTP

b. HTTP

c. HTTPS

d. SMTP

Correct Answer: c. HTTPS

Q10. Which method of NetworkInterface is used to check whether a network interface is up?

a. isActive()

b. isUp()

c. isRunning()

d. isConnected()

Correct Answer: b. isUp()

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ASSIGNMENT — 11

JDBC

Q1. How do you establish a connection to a database using JDBC?

a. By creating an instance of the Connection interface

b. By using the DriverManager.getConnection() method

c. By implementing the Connection interface

d. By extending the Connection class

Correct Answer: b. By using the DriverManager.getConnection() method

Q2. Which method executes a simple query and returns a single ResultSet object?

a. executeQuery()

b. executeUpdate()

c. execute()

d. run()

Correct Answer: a. executeQuery()

Q3. Which of the following ensures that the correct driver is used to access each data source?

a. java.sql.Manager

b. java.sql.DriverManager

c. java.sql.Statement

d. java.sql.DriverSession

Correct Answer: b. java.sql.DriverManager

Q4. What is the purpose of the ResultSet interface in JDBC?

a. To store the result of a query

b. To execute SQL queries

c. To manage database connections

d. To update data in the database

Answer: a. To store the result of a query

Q5. What does setAutoCommit(false) do?

a. Commits transaction after each query

b. Explicitly commits transaction

c. Does not commit transaction automatically after each query

d. Never commits transaction

Answer: c. Does not commit transaction automatically after each query

Q6. The following is a statement in Java using JDBC. Which of the following statements is FALSE?

Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/nptel", "java", "joy");

a. 3306 is the default MySQL port.

b. Database name is 'nptel'.

c. The database server is hosted on IP 127.0.0.1.

d. Username is 'java' and the password is 'joy'.

Answer: d. Username is 'java' and the password is 'joy'.

Q7. What is the correct order to close database resources?

a. Connection then Statement then ResultSet

b. ResultSet then Statement then Connection

c. Statement then Connection then ResultSet

d. Statement then ResultSet then Connection

Answer: b. ResultSet then Statement then Connection

Q8. What is the full form of JDBC?

a. Java Database Connectivity

b. Java Data Code

c. Java Data Communication

d. Java Development Connectivity

Answer: a. Java Database Connectivity

Q9. Fill in the missing code to establish a connection to a MySQL database.

a. DriverManager.connect("mysql:localhost:mydb", "user", "password");

b. DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");

c. Connection.get("jdbc:mysql://localhost:3306/mydb", "user", "password");

d. Driver.connect("jdbc:mysql://localhost:mydb", "user", "password");

Answer: b. DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");

Q10. What is the purpose of the DriverManager class in JDBC?

a. To manage database connections.

b. To execute SQL queries.

c. To fetch data from a database.

d. To represent a database record.

Answer: a. To manage database connections.

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ASSIGNMENT — 12

Q1. Which method returns the number of characters in a Java String?

a. size()

b. length()

c. count()

d. getSize()

Answer: b. length()

Q2. Which method is used to get a character at a specified position in a String?

a. charAt()

b. getChar()

c. characterAt()

d. getAt()

Answer: a. charAt()

Q3. Which keyword is used to create a subclass in Java?

a. implements

b. inherits

c. extends

d. super

Answer: c. extends

Q4. Which block is executed whether an exception occurs or not?

a. try

b. catch

c. throw

d. finally

Answer: d. finally

Q5. Which keyword is used to explicitly throw an exception in Java?

a. throws

b. throw

c. exception

d. catch

Answer: b. throw

Q6. Which keyword is used to indicate that a method may throw an exception?

a. throw

b. throws

c. try

d. finally

Answer: b. throws

Q7. Which syntax is used to declare an array variable in Java?

a. array

b. []array

c. []

d. array[]

Answer: d. array[]

Q8. Which layout manager arranges components in five regions?

a. FlowLayout

b. BorderLayout

c. GridLayout

d. BoxLayout

Answer: b. BorderLayout

Q9. Which method is used to register an ActionListener with a JButton?

a. addActionListener()

b. addButtonListener()

c. setAction()

d. registerAction()

Answer: a. addActionListener()

Q10. Which method of NetworkInterface is used to check whether a network interface is up?

a. isActive()

b. isUp()

c. isRunning()

d. isConnected()

Answer: b. isUp()